

**Block 1 — Input protection, rectification, and 5 V linear**

| Ref | Part Number        | Description                                      | Qty |
|-----|--------------------|--------------------------------------------------|-----|
| J1  | PJ-063AH           | DC barrel jack, through-hole, center-positive    | 1   |
| F1  | OPTF0078P          | Fuseholder                                       | 1   |
| F2  | 0239003.MXP        | 3 A fuse, time-delay (use 0239003.HXP footprint) | 1   |
| D1  | SBR3U60P1          | 3 A, 60 V SBR rectifier                          | 1   |
| C1  | EDK476MD5DS9HAA    | 47 $\mu$ F electrolytic capacitor                | 1   |
| C2  | GRM21BR61H475KE51L | 4.7 $\mu$ F, 1206, X5R ceramic capacitor         | 1   |
| M1  | LM340MPX-5.0/NOPB  | 5 V linear regulator (LM7805-type)               | 1   |
| C3  | C0603C104K5RACTU   | 0.1 $\mu$ F, 0603, X7R ceramic capacitor         | 1   |
| C4  | C0603C104K5RACTU   | 0.1 $\mu$ F, 0603, X7R ceramic capacitor         | 1   |
| D2  | TL431BQDBZRQ1      | Precision shunt regulator                        | 1   |

**Block 2 — VIN sense, comparator/EN/F2 logic network**

| Ref    | Part Number      | Description                                             | Qty |
|--------|------------------|---------------------------------------------------------|-----|
| C5     | C1210C226M5PACTU | 22 $\mu$ F, 1210, X5R, 50 V ceramic capacitor           | 1   |
| R1     | CRCW060310K0FKEA | 10 k $\Omega$ , 0603, 1% resistor (primary shunt sense) | 1   |
| R2     | CRCW060310K0FKEA | 10 k $\Omega$ , 0603, 1% resistor                       | 1   |
| R3     | CRCW060310K0FKEA | 10 k $\Omega$ , 0603, 1% resistor                       | 1   |
| R5     | CRCW060310K0FKEA | 10 k $\Omega$ , 0603, 1% resistor                       | 1   |
| R7     | CRCW060310K0FKEA | 10 k $\Omega$ , 0603, 1% resistor                       | 1   |
| R1_EN1 | CRCW060310K0FKEA | 10 k $\Omega$ , 0603, 1% resistor (EN logic)            | 1   |
| R1_EN2 | CRCW060310K0FKEA | 10 k $\Omega$ , 0603, 1% resistor (EN logic)            | 1   |
| R1_EN3 | CRCW060310K0FKEA | 10 k $\Omega$ , 0603, 1% resistor (EN logic)            | 1   |
| R1_F2  | CRCW060310K0FKEA | 10 k $\Omega$ , 0603, 1% resistor (F2 logic)            | 1   |
| R2_F2  | CRCW060310K0FKEA | 10 k $\Omega$ , 0603, 1% resistor (F2 logic)            | 1   |
| R4     | CRCW060327K4FKEA | 27.4 k $\Omega$ , 0603, 1% resistor                     | 1   |
| R6     | CRCW060339K2FKEA | 39.2 k $\Omega$ , 0603, 1% resistor                     | 1   |

**Block 3 — Logic switching, comparator, P-FET branch**

| Ref   | Part Number     | Description                                   | Qty |
|-------|-----------------|-----------------------------------------------|-----|
| C1_L1 | C1210C226M8PACT | 22 $\mu$ F, 1210, X6S, 63 V ceramic capacitor | 1   |

|              |                  |                                                                       |   |
|--------------|------------------|-----------------------------------------------------------------------|---|
|              | U                |                                                                       |   |
| <b>C2_L1</b> | C1210C226M8PACTU | 22 $\mu$ F, 1210, X6S, 63 V ceramic capacitor                         | 1 |
| <b>L1</b>    | LM339BIRTER      | Quad comparator, open-collector outputs                               | 1 |
| <b>T1</b>    | FMB3904          | NPN transistor, SOT-23                                                | 1 |
| <b>R7</b>    | CRCW0603100KFKEA | 100 k $\Omega$ , 0603, 1% resistor (VIN_FUSED branch to P-FET source) | 1 |
| <b>M2</b>    | SI2333CDS-T1-GE3 | P-channel MOSFET, logic-level, SOT-23                                 | 1 |

#### Block 4 — Buck1 (TPS543321) + peripherals (updated R9)

| Ref            | Part Number        | Description                                                          | Qty |
|----------------|--------------------|----------------------------------------------------------------------|-----|
| <b>C6</b>      | C1210C226M5PACTU   | 22 $\mu$ F, 1210, X5R, 50 V ceramic capacitor (input bulk)           | 1   |
| <b>C7</b>      | C0603C104K5RACTU   | 0.1 $\mu$ F, 0603, X5R, 50 V ceramic capacitor (input bypass)        | 1   |
| <b>U1</b>      | TPS543321RQFR      | 3 A synchronous buck regulator                                       | 1   |
| <b>L1</b>      | IHLP2525CZER2R2M01 | 2.2 $\mu$ H shielded inductor, 2525 footprint                        | 1   |
| <b>R8</b>      | TNPV1206330KBYEA   | 330 k $\Omega$ , 1206, 0.1% resistor (MODE pin configuration)        | 1   |
| <b>Boot_C1</b> | C0402C223K5RACTU   | 22 nF, 0402, X7R, 50 V ceramic capacitor (bootstrap)                 | 1   |
| <b>R9</b>      | CRCW040256K2FKED   | 56.2 k $\Omega$ , 0402, 1% resistor (feedback top, raised to ~5.3 V) | 1   |
| <b>R10</b>     | CRCW060310K0FKEA   | 10 k $\Omega$ , 0603, 1% resistor (feedback bottom)                  | 1   |
| <b>C8</b>      | C1210C226M5PACTU   | 22 $\mu$ F, 1210, X5R, 50 V ceramic capacitor (output bulk)          | 1   |
| <b>C9</b>      | C0603C104K5RACTU   | 0.1 $\mu$ F, 0603, X5R, 50 V ceramic capacitor (output bypass)       | 1   |

#### Block 5 — Buck2 (TPS563300) + peripherals (updated R10)

| Ref            | Part Number        | Description                                                          | Qty |
|----------------|--------------------|----------------------------------------------------------------------|-----|
| <b>C10</b>     | C1210C226M5PACTU   | 22 $\mu$ F, 1210, X5R, 50 V ceramic capacitor (input bulk)           | 1   |
| <b>C11</b>     | C0603C104K5RACTU   | 0.1 $\mu$ F, 0603, X5R, 50 V ceramic capacitor (input bypass)        | 1   |
| <b>U2</b>      | TPS563300DRLR      | 3 A synchronous buck regulator                                       | 1   |
| <b>L2</b>      | IHLP2525CZER2R2M01 | 2.2 $\mu$ H shielded inductor, 2525 footprint                        | 1   |
| <b>Boot_C2</b> | C0402C223K5RACTU   | 22 nF, 0402, X7R, 50 V ceramic capacitor (bootstrap)                 | 1   |
| <b>R10</b>     | CRCW040256K2FKED   | 56.2 k $\Omega$ , 0402, 1% resistor (feedback top, raised to ~5.3 V) | 1   |
| <b>R11</b>     | CRCW060310K0FKEA   | 10 k $\Omega$ , 0603, 1% resistor (feedback bottom)                  | 1   |
| <b>C12</b>     | C1210C226M5PACTU   | 22 $\mu$ F, 1210, X5R, 50 V ceramic capacitor (output bulk)          | 1   |
| <b>C13</b>     | C0603C104K5RACTU   | 0.1 $\mu$ F, 0603, X5R, 50 V ceramic capacitor (output bypass)       | 1   |

#### Block 6 — Buck3 (LT8609A) + peripherals (updated R12/R13)

| Ref            | Part Number        | Description                                                        | Qty |
|----------------|--------------------|--------------------------------------------------------------------|-----|
| <b>C10</b>     | C1210C226M5PACTU   | 22 $\mu$ F, 1210, X5R, 50 V ceramic capacitor (input bulk)         | 1   |
| <b>C11</b>     | C0603C104K5RACTU   | 0.1 $\mu$ F, 0603, X5R, 50 V ceramic capacitor (input bypass)      | 1   |
| <b>U3</b>      | LT8609AEDDM#TRPBF  | 3 A, 2 MHz synchronous buck regulator                              | 1   |
| <b>C12</b>     | C0805C105J5RACTU   | 1 $\mu$ F, 0805, C0G, 50 V ceramic capacitor (INTVcc decoupling)   | 1   |
| <b>R11</b>     | CRCW060318K2FKEA   | 18.2 k $\Omega$ , 0603, 1% resistor                                | 1   |
| <b>Boot_C3</b> | C0603C104K5RACTU   | 0.1 $\mu$ F, 0603, X5R, 50 V ceramic capacitor (bootstrap)         | 1   |
| <b>L2</b>      | IHLP2525CZER2R2M01 | 2.2 $\mu$ H shielded inductor, 2525 footprint                      | 1   |
| <b>R12</b>     | CRCW0603845KFKEA   | 845 k $\Omega$ , 0603, 1% resistor (feedback top, targets ~5.36 V) | 1   |
| <b>R13</b>     | CRCW0603187KFKEA   | 187 k $\Omega$ , 0603, 1% resistor (feedback bottom)               | 1   |
| <b>C13</b>     | 600S100JT250XT     | 10 pF, 250 V RF ceramic capacitor (compensation)                   | 1   |
| <b>C14</b>     | C1210C226M5PACTU   | 22 $\mu$ F, 1210, X5R, 50 V ceramic capacitor (output bulk)        | 1   |
| <b>C15</b>     | C0603C104K5RACTU   | 0.1 $\mu$ F, 0603, X5R, 50 V ceramic capacitor (output bypass)     | 1   |

### Block 7 — TPS\_1 (Buck1 5V\_OUT protection)

| Ref            | Part Number        | Description                                                      | Qty |
|----------------|--------------------|------------------------------------------------------------------|-----|
| <b>TPS_1</b>   | TPS1H200AQDGNRQ1   | Smart high-side switch, 40 V, 200 m $\Omega$                     | 1   |
| <b>R_CL_1</b>  | CRCW06031K20FKEA   | 1.2 k $\Omega$ , 0603, 1% resistor (current-limit configuration) | 1   |
| <b>C_DEL_1</b> | VJ0805Y681KXACW1BC | 680 pF, 0805, Y5V ceramic capacitor (delay/timing)               | 1   |
| <b>C16</b>     | VJ0603Y105KXACW1BC | 1 $\mu$ F, 0603, Y5V ceramic capacitor (decoupling)              | 1   |
| <b>C17</b>     | VJ0603Y104KXXAC    | 0.1 $\mu$ F, 0603, X7R ceramic capacitor (bypass/filter)         | 1   |

### Block 8 — TPS\_2 (Buck2 5V\_OUT protection)

| Ref            | Part Number        | Description                                                      | Qty |
|----------------|--------------------|------------------------------------------------------------------|-----|
| <b>TPS_2</b>   | TPS1H200AQDGNRQ1   | Smart high-side switch, 40 V, 200 m $\Omega$                     | 1   |
| <b>R_CL_2</b>  | CRCW06031K20FKEA   | 1.2 k $\Omega$ , 0603, 1% resistor (current-limit configuration) | 1   |
| <b>C_DEL_2</b> | VJ0805Y681KXACW1BC | 680 pF, 0805, Y5V ceramic capacitor (delay/timing)               | 1   |
| <b>C18</b>     | VJ0603Y105KXACW1BC | 1 $\mu$ F, 0603, Y5V ceramic capacitor (decoupling)              | 1   |
| <b>C19</b>     | VJ0603Y104KXXAC    | 0.1 $\mu$ F, 0603, X7R ceramic capacitor (bypass/filter)         | 1   |

### Block 9 — TPS\_3 (Buck3 5V\_OUT protection)

| Ref           | Part Number      | Description                                                      | Qty |
|---------------|------------------|------------------------------------------------------------------|-----|
| <b>TPS_3</b>  | TPS1H200AQDGNRQ1 | Smart high-side switch, 40 V, 200 m $\Omega$                     | 1   |
| <b>R_CL_3</b> | CRCW06031K20FKEA | 1.2 k $\Omega$ , 0603, 1% resistor (current-limit configuration) | 1   |

|                |                    |                                                          |   |
|----------------|--------------------|----------------------------------------------------------|---|
| <b>C_DEL_3</b> | VJ0805Y681KXACW1BC | 680 pF, 0805, Y5V ceramic capacitor (delay/timing)       | 1 |
| <b>C20</b>     | VJ0603Y105KXACW1BC | 1 $\mu$ F, 0603, Y5V ceramic capacitor (decoupling)      | 1 |
| <b>C21</b>     | VJ0603Y104KXXAC    | 0.1 $\mu$ F, 0603, X7R ceramic capacitor (bypass/filter) | 1 |

## Block 10 — Shared 5V\_OUT node

| Ref              | Part Number        | Description                                                                | Qty |
|------------------|--------------------|----------------------------------------------------------------------------|-----|
| <b>C_BULK_5V</b> | T54EE476M050ESA100 | 47 $\mu$ F, 50 V polymer tantalum capacitor, 2917 (bulk output cap)        | 1   |
| <b>C_MF_5V</b>   | VJ0603Y105KXACW1BC | 1.0 $\mu$ F, 16 V, X7R, 0603 ceramic capacitor (connector-side, mid-freq)  | 1   |
| <b>C_HF_5V</b>   | VJ0603Y104KXXAC    | 0.1 $\mu$ F, 25 V, X7R, 0603 ceramic capacitor (connector-side, high-freq) | 1   |
| <b>D_TVS_5V</b>  | SMF5.0A            | 5 V TVS diode, SOD-123FL                                                   | 1   |
| <b>J_USB_C</b>   | USB4125-GF-A       | USB-C receptacle, mid-mount                                                | 1   |

## Block 11 — External adapter plug + recommended 1 m cables

| Ref                | Part Number     | Description                                                             | Qty |
|--------------------|-----------------|-------------------------------------------------------------------------|-----|
| <b>P_ADAPT</b>     | 70235K607       | Barrel plug for 12 V, 2 A adapter (mates with PJ-063AH)                 | 1   |
| <b>C_USBCC_1M</b>  | CAB004BT1MWH    | Belkin 1 m USB-C to USB-C cable, 60 W-class, 3 A-rated power conductors | 1   |
| <b>C_USBA_C_1M</b> | F2CU029BT1M-BLK | Belkin 1 m USB-A to USB-C cable, up to 3 A charging                     | 1   |
| <b>C_USBA_L_1M</b> | CAA001BT1MWH    | Belkin 1 m USB-A to Lightning cable, MFi-class for iPhone/iPad          | 1   |